HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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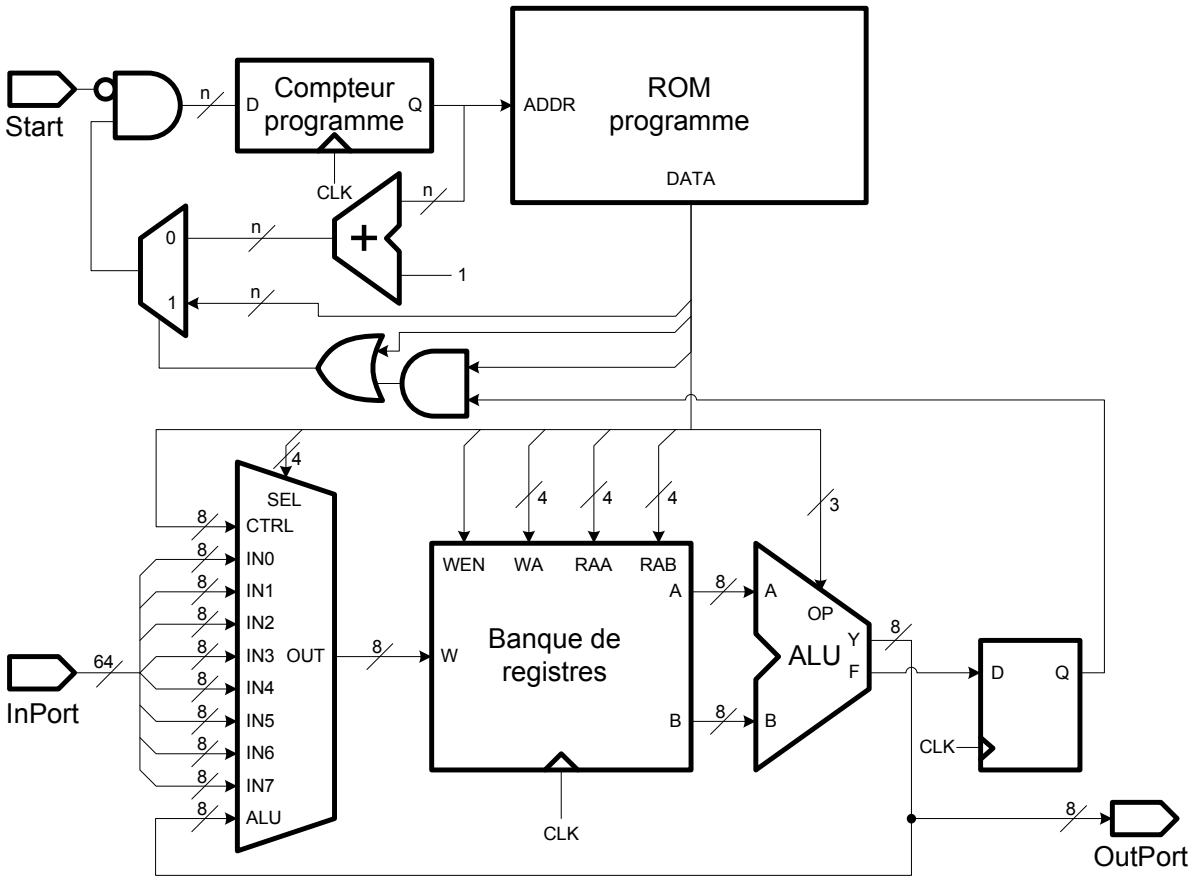
Formal verification
Verification of *small* microprocessor
CERN training - November 2026

Context

A team has been developing a micro-coded processor, able to count the number of 1s found on an input bus of 64 bits. This processor is decomposed into various modules and we would like to formally verify this design.
Let's dispatch the work onto the team.

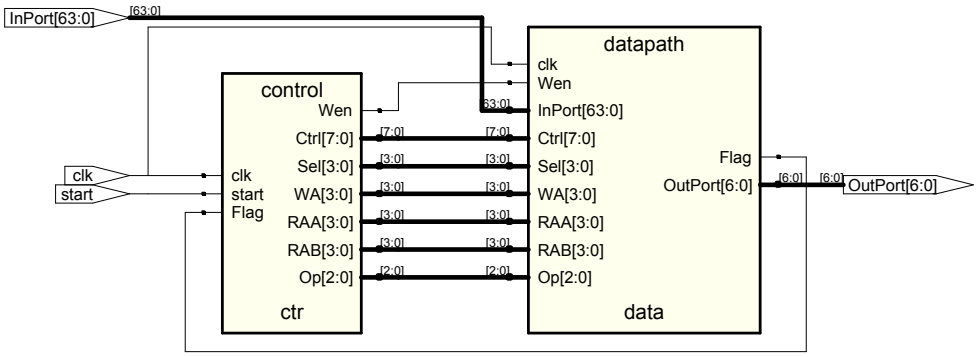
1 Full processor

The general architecture is the following [1] :



1. Yes, there is some French on the figures, but I guess you'll figure out what's happening.

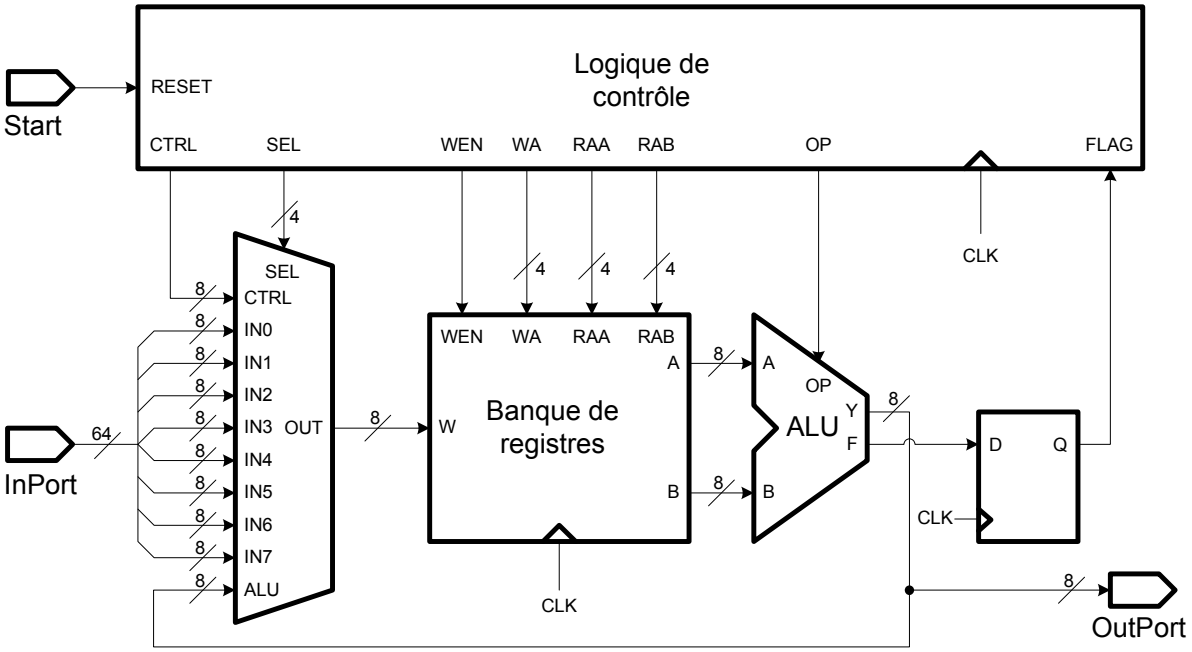
It is however decomposed into a control and a datapath modules :



The control part is itself decomposed into a program counter and a ROM memory.

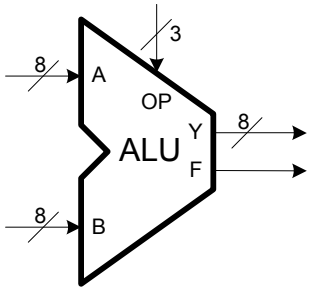
2 Datapath

The datapath is decomposed into a bank register, an Arithmetic Logic Unit (ALU) and the rest of the parts (mux + D latch).
The datapath corresponds to the low part of the following schematic :



3 Arithmetic Logic Unit

The ALU does offer the following interface :



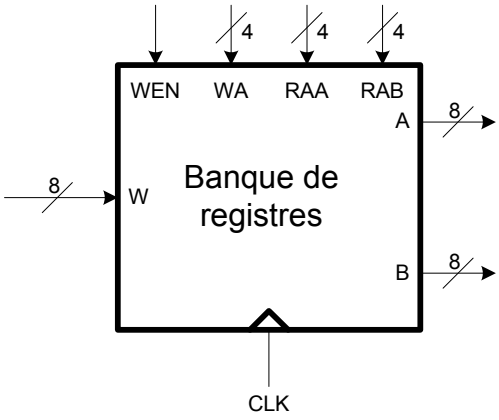
Depending on the `OP`, the operation is the following :

Operation	OP	Y output	F output
ADD	000	$A + B$	–
SHR	001	$A \gg 1$	–
EQ	010	–	$A = B$
AND	011	$A \; and \; B$	–
MOV	100	A	–

An em dash means that output is not specified for that operation.

4 Register bank

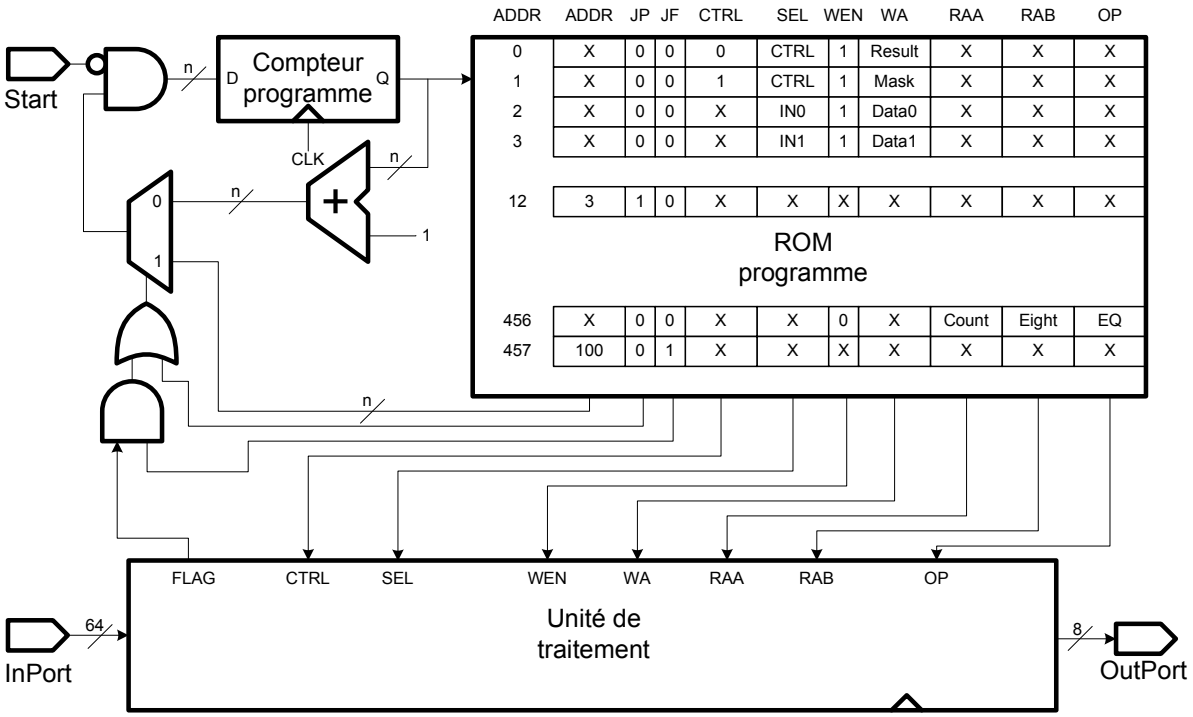
The register bank corresponds to the following module :



It contains 16 8-bit elements. Two read ports, `A` and `B`, allow simultaneous reads from addresses `RAA` and `RAB`, respectively. Reads are asynchronous. Writes are synchronous and enabled by `WEN`; when enabled, data `W` is written to address `WA`.

5 Program counter

The program counter increments by default. When `start` is high (`'1'`), it is synchronously set to 0. Otherwise, when `Flag` and `JF` are both active, it loads the supplied address. It also loads the supplied address when `JP` is active.



6 Modules to verify

The following modules can be individually verified :

1. Program counter
2. ALU
3. Register bank
4. Datapath
5. Everything except the ROM
6. The entire block (the ROM contains a program to count the number of '1's)